# ICAD RT-5 · Set 8 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

## Physics

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| 10 | a | b | c | d |
| 11 | a | b | c | d |
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## Chemistry

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*Numerical — write the value*

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## Mathematics

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| 13 | a | b | c | d |
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*Numerical — write the value*

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*Bubbles — single correct · Box — write the value · each subject out of 100*

# ICAD RT-5 · Set 8

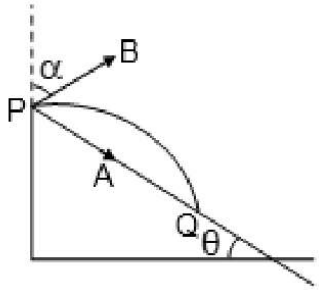
Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

## Physics

**PART A · Q1–20 · SINGLE CORRECT**

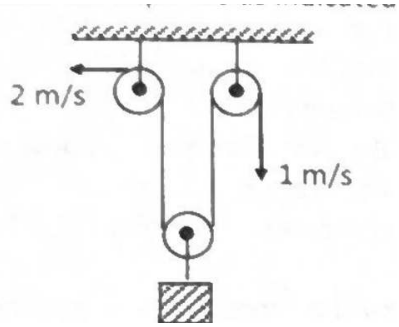
**1.** P4L4V3Q3 A body is projected vertically upwards with a velocity of 10 $ms^{-1}$. It reaches the maximum vertical height h in time t. In t/2, the height covered is: (a) h/2 (b) 2h/5 (c) 3h/4 (d) 5h/8

**2.** P4XVRT3Q6 The particle A is projected from point P with velocity 'u' along the plane and simultaneously another particle B with velocity 'v' at an angle 'α' with vertical. The particles collide at point Q on the plane. Then : (a) $-v \sin(\theta - \alpha) = u$ (b) $v \cos(\theta - \alpha) = u$ (c) $v = u$ (d) $\tan(\theta - \alpha) = u$

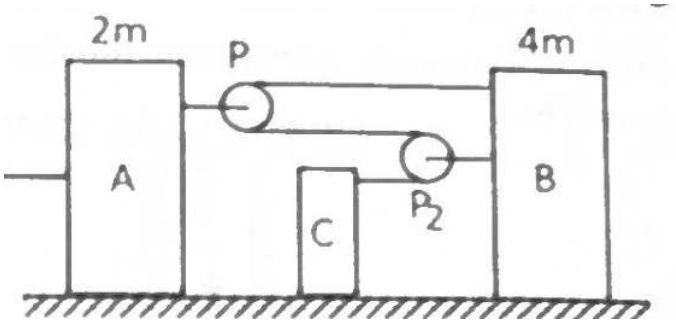



**3.** P5L2V2Q4 What is the tension in a massless string with 10 N force applied (at each end)? (a) 10 N (b) 5 N (c) 0 N (d) Depends on length

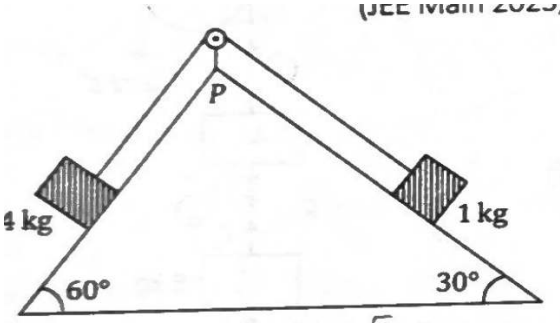
**4.** P6L3V6Q4 Find the velocity of the hanging block if the velocities of the free ends of the rope are as indicated in the figure (the rope passes over two fixed pulleys and under the movable pulley carrying the block; its left free end is pulled horizontally away at 2 m/s and its right free end moves down at 1 m/s). (a) (5/2) m/s ↑ (b) (5/2) m/s ↓ (c) (1/2) m/s ↑ (d) (3/2) m/s ↑



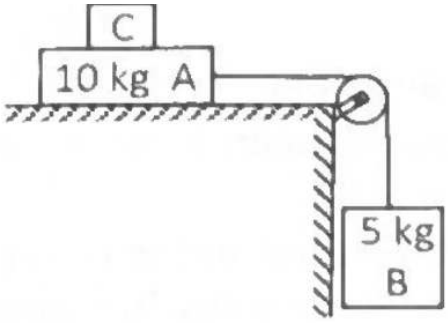

**5.** P6L4V6Q6 Block A (2m) and block B (4m) rest on a smooth floor; F pulls A to the left. One light string is tied to B, runs left round pulley P (attached to A), back right round pulley $P_2$ (attached to B) and then left to block C of mass m, which also rests on the smooth floor (in the previous question this end was tied to a fixed support). All surfaces smooth, pulleys and string light. Find the acceleration of block B. (a) $a_B = 3F/(20m)$ m/s² (b) $a_B = 3F/(21m)$ m/s² (c) $a_B = 2F/(21m)$ m/s² (d) $a_B = 3F/(18m)$ m/s²



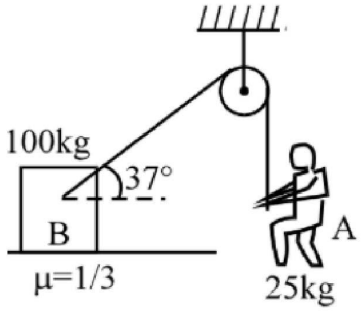

**6.** P6SBVPYQQ6 As per given figure, a weightless pulley P is attached on a double inclined frictionless surface (4 kg block on the 60° face, 1 kg block on the 30° face, joined by a string over P). The tension in the string (massless) will be (if g = 10 m $s^{-2}$) (a) $(4\sqrt{3} + 1)$ N (b) $4(\sqrt{3} + 1)$ N (c) $4(\sqrt{3} - 1)$ N (d) $(4\sqrt{3} - 1)$ N




**7.** P7L4V4Q4 Masses A (10 kg) and B (5 kg) are connected by a string over a frictionless pulley at a table corner; A lies on the table with μ = 0.2 and B hangs. The minimum mass of C that may be placed on A to prevent it from moving is: (a) 15 kg (b) 10 kg (c) 5 kg (d) zero



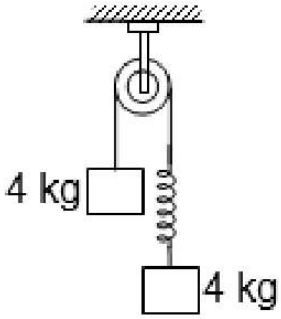
**8.** P7XVRT3Q16 Block B of mass 100 kg rests on a rough surface of friction coefficient μ = 1/3. A rope is tied to block B as shown in figure. The maximum acceleration with which boy A of 25 kg can climb on rope without making block move is : (a) 4g/3 (b) g/3 (c) g/2 (d) 3g/4



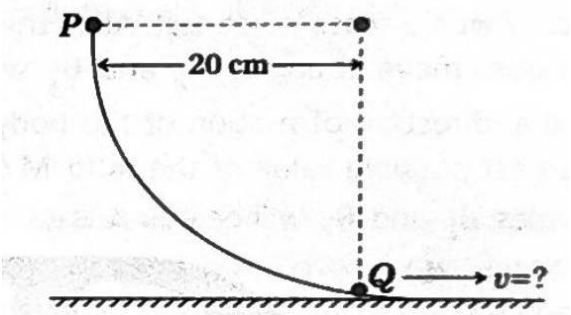

**9.** P8L2V2Q5 A force F⃗ = (5î + 3ĵ + 2k̂) N is applied over a particle which displaces it from its origin to the point r⃗ = (2î − ĵ) m. The work done on the particle in joules is (a) −7 (b) +7 (c) +10 (d) +13

**10.** P8L6VSuggQ2 Calculate gravitational potential energy of a 2 kg object raised 5 m above ground. (Use g = 10 m/s²) (a) 20 J (b) 50 J (c) 100 J (d) 10 J

**11.** P8XVCAT5Q8 In figure spring constant of the spring is 100 N/m. Extension produced in the spring is : (g = 10 m/s²) (a) 10 cm (b) 20 cm (c) 30 cm (d) 40 cm




**12.** P9L2V6Q1 As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assuming the collision to be elastic, the velocity of ball Q after collision will be (g = 10 m/s², radius/horizontal run shown = 20 cm) (a) 4 m/s (b) 0 (c) 2 m/s (d) 0.25 m/s




**PART B · Q21–25 · NUMERICAL VALUE**

**21.** P3SBVPYQQ2 Two particles at equal distance from the origin, Ā = 2î + 3nĵ + 2k̂ and B̄ = 2î − 2ĵ + 4pk̂ ; both vectors at right angles — find $n^{-1}$

**22.** P4L6VSuggQ3 Between two stations, a train accelerates uniformly, moves at constant speed, then retards uniformly. Ratio of times = 1:8:1, greatest speed = 60 km $h^{-1}$. Find the average speed over the whole journey

**13.** P9L4V6Q3 What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass? (Assume the collision to be head-on elastic collision) (a) 50.0% (b) 66.6% (c) 55.5% (d) 33.3%

**14.** P9SBVPYQQ4 A body of mass 1000 kg is moving horizontally with a velocity 6 m $s^{-1}$ . If 200 kg extra mass is added, the final velocity (in m $s^{-1}$ ) is: (a) 6 (b) 2 (c) 3 (d) 5

**15.** P10L1VSuggQ3 A wheel starts rotating with an angular velocity of 10π rad/s and undergoes constant angular acceleration. After 8 seconds, its angular velocity becomes 30π rad/s. How many complete revolutions does the wheel make during this interval? (a) 80 (b) 120 (c) 160 (d) 200

**16.** P10L3V4Q3 An object is moving on a circular track of radius 450 m. At some instant, the object is moving at 30 m/s and gaining speed at a uniform rate of 2 $m/s^2$. Its acceleration at this instant is nearly: (a) 1.6 $m/s^2$ (b) 0.8 $m/s^2$ (c) 2.8 $m/s^2$ (d) 3.4 $m/s^2$

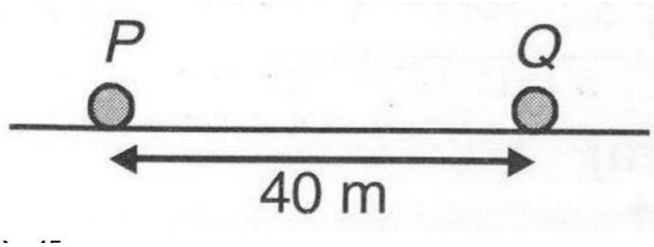
**17.** P10L4V5Q4 A mass 3 kg is suspended from a massless string of length 50 cm. Initially, the mass is at rest with the string along the vertical. Another object of mass 200 g moving horizontally at a speed 100 m/s hits the suspended body and sticks to it, then: (a) they are able to complete vertical circle (b) they are unable to complete vertical circle (c) their system begins to oscillate about the original position of 3 kg mass (d) tension in the string remains constant

**18.** P11L1V2Q5 The distance between the carbon atom and the oxygen atom in a carbon monoxide molecule is 1.1 Å. Given, mass of carbon atom is 12 a.m.u. and mass of oxygen atom is 16 a.m.u., calculate the position of the center of mass of the carbon monoxide molecule (a) 6.3 Å from the carbon atom (b) 1 Å from the oxygen atom (c) 0.63 Å from the carbon atom (d) 0.12 Å from the oxygen atom

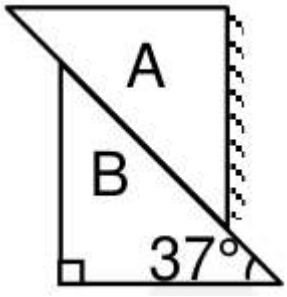
**19.** P11L2V4Q1 The linear mass density (λ) of a rod of length l kept along x-axis varies as λ = α + βx, where α and β are positive constants. The centre of mass of the rod is at (a) (2β + 3αL)L / (2(2β + αL)) (b) (3α + 2βL)L / (3(2α + βL)) (c) (3β + 2αL)L / (3(2β + αL)) (d) (3β + 2αL)L / (3β + 2α)

**20.** P11L4V4Q5 A man of mass 60 kg is standing on a boat of mass 140 kg, which is at rest in still water. The man is initially at 20 m from the shore. He starts walking on the boat for 4 s with constant speed 1.5 m/s (relative to the boat) towards the shore. The final distance of the man from the shore is (a) 15.8 m (b) 4.2 m (c) 12.6 m (d) 14.1 m

**23.** P4XVCAT3Q18 Two particles P and Q are initially 40 m apart, P behind Q. Particle P starts moving with a uniform velocity 10 m/s towards Q. Particle Q starting from rest has an acceleration 2 m/s² in the direction of velocity of P. Then the minimum distance between P and Q will be (A) 45 m (B) 15 m (C) 35 m (D) 30 m




**24.** P6XVCAT4Q24 The acceleration of block A is 5 m/s². Then acceleration of the block B, if answer = (4x/3) m/s², then x = ? (Wedge B, right angle at bottom-left, rests on the floor; block A rests on its 37° hypotenuse against a smooth vertical wall, as in the figure.)




**25.** P10XVCAT6Q23 A heavy particle hanging from a string of length l is projected horizontally with speed √(gl). The speed of the particle at the point where the tension in the string equals weight of the particle is found to be √(3gl/n). Find value of n.

## Chemistry

**PART A · Q1–20 · SINGLE CORRECT**

**1.** C1L5VSuggQ2 Under identical conditions, hydrogen diffuses 4 times faster than a gas X. Calculate the molecular mass of X. (a) 4 g $mol^{-1}$ (b) 8 g $mol^{-1}$ (c) 16 g $mol^{-1}$ (d) 32 g $mol^{-1}$

**2.** C2L4VSuggQ2 If $a_0 = 5.29 \times 10^{-11}$ m, calculate radius of the second Bohr orbit of hydrogen atom. (a) $1.06 \times 10^{-9}$ m (b) $2.12 \times 10^{-11}$ m (c) $2.12 \times 10^{-10}$ m (d) $1.06 \times 10^{-10}$ m

**3.** C2L6VSuggQ3 The time period (T) of revolution of an electron in an orbit is given by T = 1/f. For a hydrogen atom in the first Bohr orbit, if the orbital frequency is $f = 6.6 \times 10^{15}$ $s^{-1}$, then the time period T is: (a) $1.5 \times 10^{-16}$ s (b) $2.0 \times 10^{-16}$ s (c) $1.0 \times 10^{-16}$ s (d) $6.6 \times 10^{-16}$ s

**4.** C3L5VSuggQ2 How many moles of $CO_2$ are produced when 44 g of $CO_2$ is formed in the decomposition: $CaCO_3 \rightarrow CaO + CO_2$ (a) 0.5 mol (b) 1 mol (c) 2 mol (d) 4 mol

**5.** C4L2V6Q9 Using Slater's rules, calculate the shielding constant (σ) for a 3p electron in Sulfur (Z = 16). Electronic configuration of S (grouped as per Slater): $(1s^2)(2s^2\ 2p^6)(3s^2\ 3p^4)$. (a) 10.55 (b) 11.25 (c) 12.40 (d) 13.10

**6.** C4L4VSuggQ3 Ionisation enthalpy of nitrogen is more than oxygen because of (a) extra stability of half filled orbitals (b) more number of energy levels (c) less number of valence electrons (d) smaller size.

**7.** C4L6V6Q7 The electronegativity of the following elements increase in the order: (a) C, N, Si, P (b) N, Si, C, P (c) Si, P, C, N (d) P, Si, N, C

**8.** C4XVRT4Q43 Ionization energies of five elements in kcal/mol are given below (table: Atom | I | II | III — P 300 549 920; Q 99 734 1100; R 118 1091 1652; S 176 347 1848; T 497 947 1500). If Q reacts with fluorine and oxygen, the molecular formula of fluoride and oxide will be respectively (a) QF, $Q_2O$ (b) $QF_2$, QO (c) $QF_3$, $Q_2O_3$ (d) $QF_4$, $QO_2$

| Atom | I | II | III |
|---|---|---|---|
| P | 300 | 549 | 920 |
| Q | 99 | 734 | 1100 |
| R | 118 | 1091 | 1652 |
| S | 176 | 347 | 1848 |
| T | 497 | 947 | 1500 |

**9.** C5L4V4Q10 On hybridization of one s- and one p-orbital , we get: (a) two mutually perpendicular orbitals (b) two orbitals at 180° (c) four orbitals directed tetrahedrally (d) three orbitals in the plane

**10.** C5L5VSuggQ2 Which molecule is T or arrow (→) shaped? (a) $BeF_2$ (b) $BCl_3$ (c) $NH_3$ (d) $ClF_3$

**11.** C5L9V5Q3 What is the correct dipole moment of $NH_3$ and $NF_3$ respectively? (a) $4.90 \times 10^{-30}$ C·m and $0.80 \times 10^{-30}$ C·m (b) $0.80 \times 10^{-30}$ C·m and $4.90 \times 10^{-30}$ C·m (c) $4.90 \times 10^{-30}$ C·m and $4.90 \times 10^{-30}$ C·m (d) $0.80 \times 10^{-30}$ C·m and $0.80 \times 10^{-30}$ C·m

**12.** C5SBVPYQQ116 Which of the following pair of molecules contain an odd electron molecule and an expanded octet molecule ? (a) BCl 3 and SF 6 (b) NO and H 2 SO 4 (c) SF 6 and H 2 SO 4 (d) BCl 3 and NO

**13.** C5XVCAT5Q43 Match Column-I with Column-II: Column-I: (P) $AsO_4^{3-}$ (Q) $ICl_2^+$ (R) $SOF_4$ (S) $XeOF_4$. Column-II: (i) All three p-orbitals used in hybridisation (ii) Tetrahedral shape (iii) Axial d-orbital with two nodal cones used in hybridisation (iv) All bond lengths are identical (v) pπ–dπ bond(s) present. (a) (P) – (i), (ii), (iv), (v); (Q) – (i), (iv); (R) – (i), (iii), (v); (S) – (i), (iii), (v) (b) (P) – (ii); (Q) – (iii); (R) – (iv); (S) – (iv) (c) (P) – (iii), (v); (Q) – (i), (v); (R) – (ii), (v); (S) – (i) (d) (P) – (v); (Q) – (ii), (iv); (R) – (ii), (iv), (v); (S) – (iii)

| | Column – I | | Column – II |
|---|---|---|---|
| (P) | $AsO_4^{3-}$ | (i) | All three p-orbitals used in hybridisation |
| (Q) | $ICl_2^+$ | (ii) | Tetrahedral shape |
| (R) | $SOF_4$ | (iii) | Axial d-orbital with two nodal cones used in hybridisation |
| (S) | $XeOF_4$ | (iv) | All bond lengths are identical |
| | | (v) | pπ–pπ bond(s) present |

**14.** C6L3V4Q10 Calculate the maximum work when 24 g of $O_2$ are expanded isothermally and reversibly from the pressure of 1.6 bar to 1.0 bar at 298 K. (a) 0.67 kJ (b) 1.87 kJ (c) 0.87 kJ (d) 1.67 kJ

**15.** C6L4V4Q10 An exothermic reaction is one in which the reacting substance: (a) have more energy than the product (b) have less energy than the products (c) have as much energy as the products (d) are at a higher temperature than the products

**16.** C6L6VSuggQ3 Which of the following is correct? (a) C_V = (∂U/∂T)_P (b) C_P = (∂H/∂T)_V (c) C_P − C_V = R (d) (∂U/∂V)_T = −a/V²

**17.** C6L7V6Q9 For complete combustion of ethanol, C 2 H 5 OH( l ) + 3O 2 (g) → 2CO 2 (g) + 3H 2 O( l ), the amount of heat produced as measured in bomb calorimeter is 1364.47 kJ mol −1 at 25°C. Assuming ideality the Enthalpy of combustion, Δ C H, for the reaction will be: (R = 8.314 J K −1 mol −1 ) (a) −1366.95 kJ mol −1 (b) −1361.95 kJ mol −1 (c) −1460.50 kJ mol −1 (d) −1350.50 kJ mol −1

**18.** C6L9V5Q1 The bond energy of H 2 is 104.3 kcal mol −1 . It means: (N = Av. no.) (a) 104.3 kcal heat is required to break N bonds in N molecule of H 2 (b) 104.3 kcal heat is required to break up N molecule to 2N atoms of H (c) 104.3 kcal heat is evolved during combination of 2N atoms of H to forms N molecule of H 2 (d) All of these

**19.** C6L12VSuggQ4 A reaction occurs spontaneously if: (a) TΔS = ΔH and both ΔH and ΔS are positive (b) TΔS > ΔH and both ΔH and ΔS are positive (c) TΔS < ΔH and both ΔH and ΔS are positive (d) TΔS > ΔH and ΔH is positive and ΔS is negative

**20.** C6SBVPYQQ94 C(s) + O 2 (g) → CO 2 (g) + 400 kJ; C(s) + (1/2)O 2 (g) → CO(g) + 100 kJ. When coal of purity 60% is allowed to burn in presence of insufficient oxygen, 60% of carbon is converted into 'CO' and the remaining is converted into 'CO 2 '. The heat generated when 0.6 kg of coal is burnt is: (a) 1600 kJ (b) 3200 kJ (c) 4400 kJ (d) 6600 kJ

### PART B · Q21–25 · NUMERICAL VALUE

**21.** C1XVRT2Q50 A volume of 100 ml of hydrogen was confined in a diffusion tube and exposed to air; at equilibrium a volume of 26 ml of air was measured in the tube. When 100 ml of an unknown gas was placed in the same tube and exposed to air, 130 ml of air was measured at equilibrium. The molecular mass of the unknown gas is

**22.** C2XVRT1Q50 The quantum number n corresponding to the excited state of $He^+$ ion, if on transition to the ground state that ion emits two photons in succession with wavelengths 108.5 nm and 30.4 nm, is ___.

**23.** C3XVCAT4Q46 A bottle of concentrated $H_2SO_4$ (density 1.80 g $cm^{-3}$) is labelled 86% by weight. What is the molarity? (round to nearest integer, in M)

**24.** C3XVRT3Q46 200 mL of 1 M HCl is mixed with 300 mL of 6 M HCl and the final solution is diluted to 1000 mL. Calculate molar concentration of [H + ] ion.

**25.** C6SBVPYQQ12 The formation enthalpies Δ f H° for H(g) and O(g) are 220.0 and 250.0 kJ mol −1 respectively at 298.15 K, and Δ f H° for H 2 O(g) is −242.0 kJ mol −1 at the same temperature. The average bond enthalpy of the O–H bond in water at 298.15 K is _____ kJ mol −1 (nearest integer).

## Mathematics

### PART A · Q1–20 · SINGLE CORRECT

**1.** M2L1VSuggQ6 If A = {1, 3, 5}, what is the cardinal number n(A)? (a) 2 (b) 3 (c) 4 (d) 5

**2.** M3SBVPYQQ6 Let α and β respectively be the maximum and the minimum values of the function $f(\theta) = 4(\sin^4(7\pi/2 - \theta) + \sin^4(11\pi + \theta)) - 2(\sin^6(3\pi/2 - \theta) + \sin^6(9\pi - \theta))$, θ ∈ R. Then α + 2β is equal to : (a) 6 (b) 4 (c) 5 (d) 3

**3.** M4L2V5Q5 f(x) = {x} + {x + 1} + {x + 2} + ... + {x + 99}, then [f(√2)] (where {·} denotes fractional part function and [·] denotes the greatest integer function) is equal to: (a) 5050 (b) 4950 (c) 41 (d) 14

**4.** M4L4VSuggQ2 The value of tan(π/4) + cot(π/4) equals: (a) 0 (b) 1 (c) 2 (d) Undefined

**5.** M5BL1VSuggQ2 Using the identity evaluate sin 75° sin 15°. (a) 1/4 (b) 1/8 (c) 1/2 (d) 0

**6.** M5CL3VSuggQ10 Given θ ∈ (0, 2π) and if Σ(r = 1 to 3) sec(rπ/6 + θ)·sec((r − 1)π/6 + θ) = 4, then which of the following alternative(s) is/are correct? (one or more correct) (a) θ = 3π/4 (b) θ = π/8 (c) θ = 7π/4 (d) θ = 3π/2

**7.** M7XVCAT4Q52 If log5120 + (x − 3) − 2log5(1 − 5^(x−3)) = −log5(0.2 − 5^(x−4)), then x is. (a) 1 (b) 2 (c) 3 (d) 4

**8.** M7XVRT4Q57 The solution of the equation log7(log5(√(x + 5) + √x)) = 0 is (are) (a) 1 (b) 3 (c) 4 (d) 5

**9.** M8ASBVPYQQ8 Consider an A.P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11 th term is (a) 90 (b) 84 (c) 122 (d) 108

**10.** M8AXVCAT5Q66 If $a_1, a_2, \ldots, a_n$ are in A.P. where $a_i > 0$ for all i, then $1/(\sqrt{a_1} + \sqrt{a_2}) + 1/(\sqrt{a_2} + \sqrt{a_3}) + \ldots + 1/(\sqrt{a_{n-1}} + \sqrt{a_n}) =$ (a) $(n-1)/(\sqrt{a_1} + \sqrt{a_n})$ (b) $(n+1)/(\sqrt{a_1} + \sqrt{a_n})$ (c) $(n-1)/(\sqrt{a_1} - \sqrt{a_n})$ (d) $(n-1)/(\sqrt{a_n} - \sqrt{a_1})$

**11.** M8BSBVPYQQ5 Let the first three terms 2, p and q, with $q \neq 2$, of a G.P. be respectively the 7 th , 8 th and 13 th terms of an A.P. If the 5 th term of the G.P. is the n th term of the A.P., then n is equal to: (a) 163 (b) 151 (c) 177 (d) 169

**12.** M8DL4V4Q9 The sum of the series $1/(\sqrt{1} + \sqrt{2}) + 1/(\sqrt{2} + \sqrt{3}) + 1/(\sqrt{3} + \sqrt{4}) + \ldots + 1/(\sqrt{(n+1)} + \sqrt{n})$ equals (a) $(2n+1)/\sqrt{n}$ (b) $(\sqrt{n} + 1)/(\sqrt{n} + \sqrt{(n-1)})$ (c) $(n + \sqrt{(n^2-1)})/(2\sqrt{n})$ (d) $\sqrt{(n+1)} - 1$

**13.** M8DL5V2Q5 Find the Geometric Mean of 4 and 16. (a) 10 (b) 8 (c) 6 (d) 12

**14.** M8DL5VSuggQ3 If a = 2 and b = 18, find G then H. [Level-3] (a) 6 and 3.6 (b) 4 and 6 (c) 8 and 7 (d) $\sqrt{36}$ and 7

**15.** M9L3V5Q4 If $(a_1 + ib_1)(a_2 + ib_2) \cdots (a_n + ib_n) = A + iB$ then $(a_1^2 + b_1^2)(a_2^2 + b_2^2) \cdots (a_n^2 + b_n^2) =$ (a) $A^2 - B^2$ (b) $A^2 + B^2$ (c) $A - B$ (d) $A + B$

**16.** M9L4V2Q2 The value of $|z - 5|$ if $z = x + iy$, is (a) $\sqrt{((x-5)^2 + y^2)}$ (b) $x^2 + \sqrt{((y-5)^2)}$ (c) $\sqrt{((x-y)^2 + 5^2)}$ (d) $\sqrt{(x^2 + (y-5)^2)}$

**17.** M9L4V3Q5 The minimum value of $|z - 1| + |z - 3|$ is (a) 4 (b) 2 (c) 0 (d) 3

**18.** M9L5V4Q4 $\sqrt{(-15 + 8i)} - \sqrt{(-15 - 8i)} =$ (a) 4i (b) 2i (c) i (d) 8i

**19.** M9L7VSuggQ13 If $i = \sqrt{(-1)}$, $\omega$ = non-real cube root of unity, then $((1+i)^{2n} - (1-i)^{2n}) / ((1 + \omega^4 - \omega^2)(1 - \omega^4 + \omega^2))$ is equal to (a) 0, if n is even (b) 0 for all $n \in I$ (c) $2^{(n-1)} \cdot i$ for all $n \in I$ (d) None of these

**20.** M9L7VSuggQ14 If $\omega$ is an imaginary cube root of unity, then the value of the expression $(2-\omega)(2-\omega^2) + 2 \cdot (3-\omega)(3-\omega^2) + \ldots\ldots + (n-1)(n-\omega)(n-\omega^2)$, is (a) $\{n(n+1)/2\}^2$ (b) $\{n(n+1)/2\}^2 - n$ (c) $\{n(n+1)/2\}^2 + n$ (d) none of these

## PART B · Q21–25 · NUMERICAL VALUE

**21.** M5AXVCAT3Q63 $a = \tan 25^\circ + \tan 35^\circ - \sqrt{3}$, $b = \cot 25^\circ + \cot 35^\circ + \sqrt{3} \Rightarrow ab =$ (A) 3 (B) 1/3 (C) −3 (D) −1/3

**22.** M6L5V5Q5 If solution set of $(x^2 + 4x + 4)/(2x^2 - x - 1) > 0$ is $(-\infty, a) \cup (a, b) \cup (c, \infty)$ then value $a - 4b + c$ is. Note: $(a < b < c)$

**23.** M6XVRT3Q74 If $||x - 3| - 4| = 1$, then the sum of values of x is

**24.** M7L3VSuggQ9 Solve the following equation (log denotes log base 10): $3^{(\log_2 \log \sqrt{x})} - \log x + \log^2 x - 3 = 0$.

**25.** M8CXVCAT6Q73 The 1st two terms of a H.P. are 2/5 and 12/23. The value of the largest positive term of the H.P. is?